

LAKSHYA

JEE 2027

Lecture No.- 28

MATHEMATICS

RELATIONS & FUNCTIONS

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Topics *to be covered*



1 Periodic Functions

2 Functional Equations

3 Question Practice

4





Question Practice

Question [IIT-JEE-2005]

79%



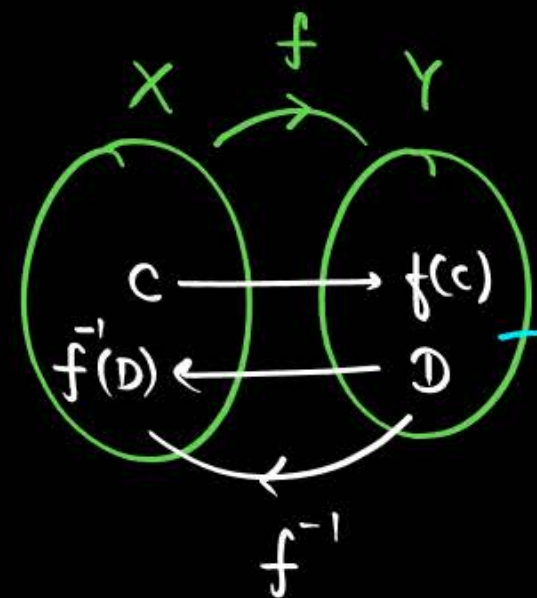
If X and Y are two non-empty sets where $f: X \rightarrow Y$, is function is defined such that $f(C) = \{f(x): x \in C\}$ for $C \subseteq X$ and $f^{-1}(D) = \{x: f(x) \in D\}$ for $D \subseteq Y$, for any $A \subseteq Y$ and $B \subseteq Y$, then

~~A~~ $f^{-1}\{f(A)\} = A$

~~B~~ $f^{-1}\{f(A)\} = A$, only if $f(X) = Y$

☒ C $f\{f^{-1}(B)\} = \textcircled{B}$, only if $B \subseteq f(x)$

~~D~~ $f\{f^{-1}(B)\} = B$



$$f^{-1}(f(x)) = x \downarrow X$$

A, B, D $f(f^{-1}(x)) = x \downarrow Y$

large.



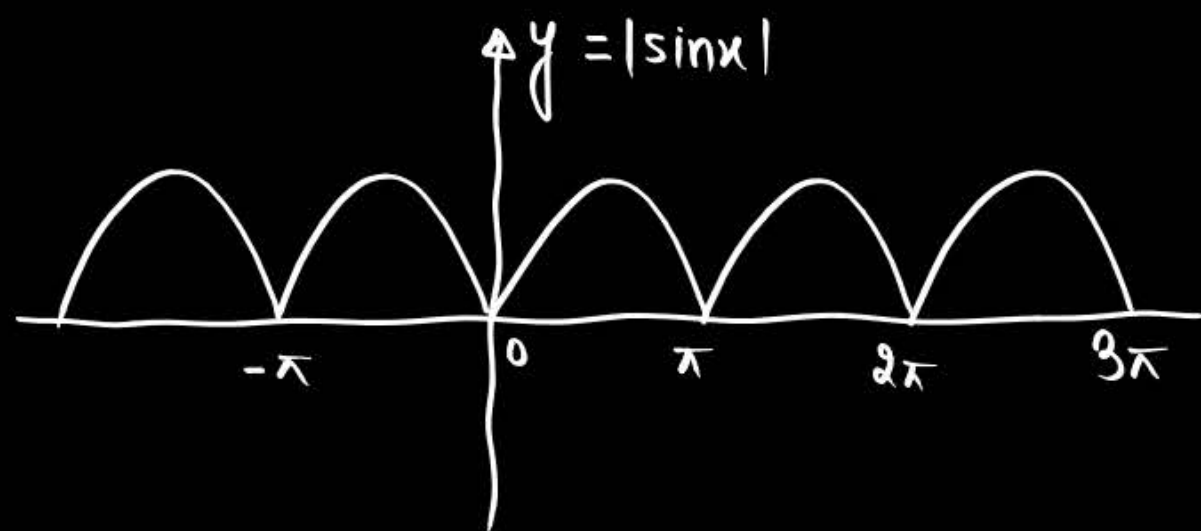
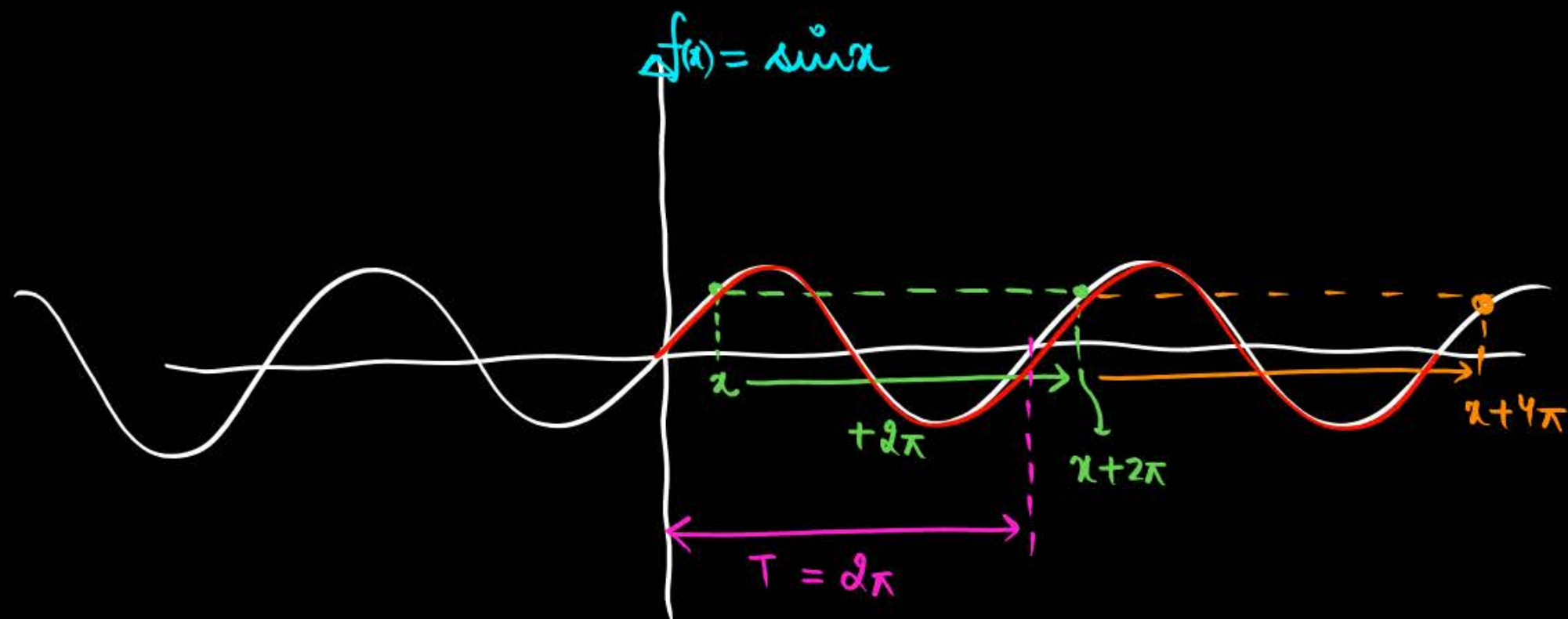
PERIODIC FUNCTIONS



$$\# \boxed{f(x+T) = f(x)} \quad \forall \underline{x} \in D_f$$

$T = \text{period of } f(x)$

$\rightarrow$ Smallest value = fundamental period.



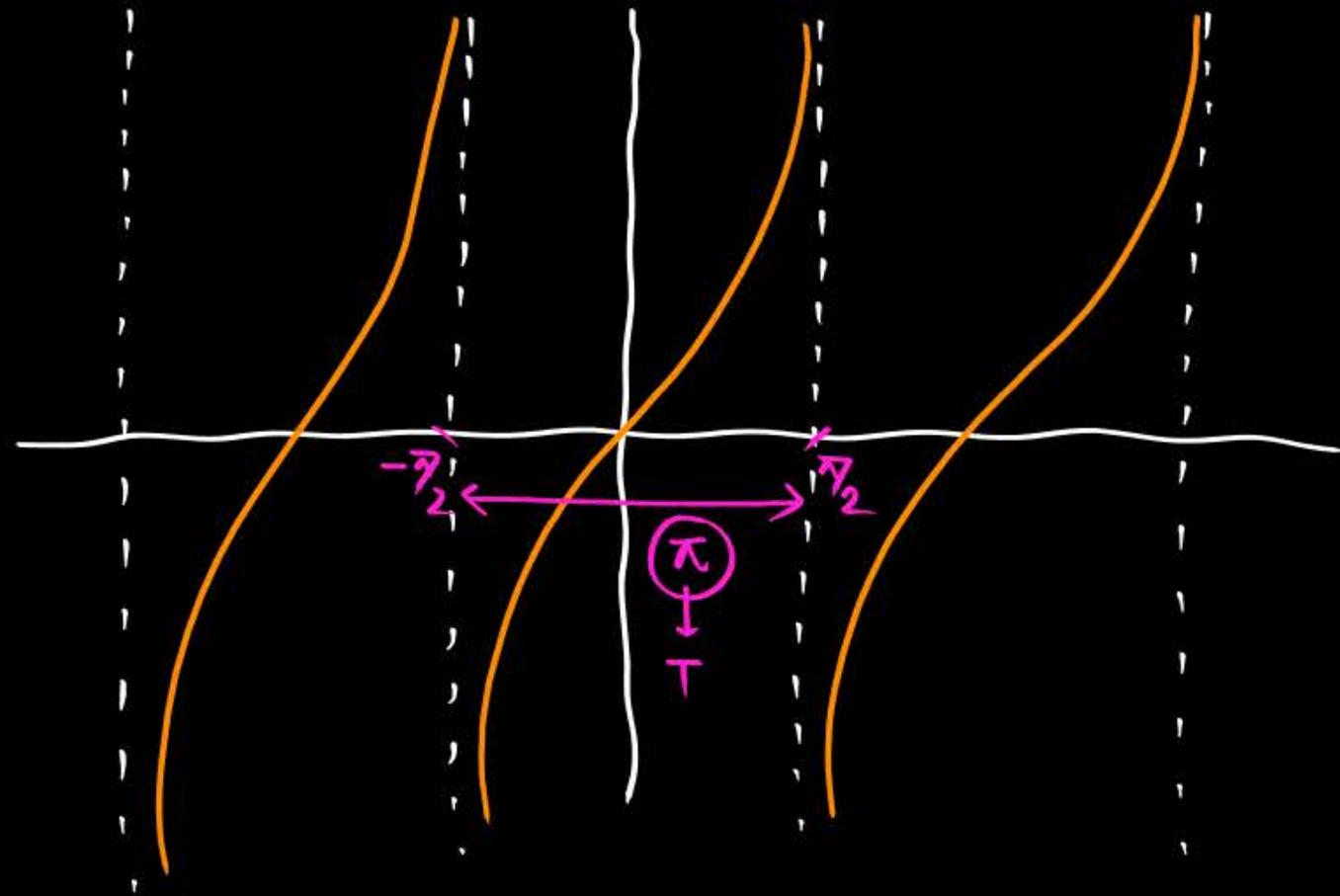
$$f(x + \overset{T}{\textcircled{2\pi}}) = f(x)$$

$$\sin(2\pi + x) = \sin(x)$$

$$\sin(\underbrace{(4\pi)}_T + x) = \sin x$$

$$\boxed{\sin(\underbrace{(\pi)}_{T \times} - x) = \sin x} \quad \times$$

$$T = \underbrace{(2\pi)}_{\text{F.P.}}, 4\pi, 6\pi, \underbrace{-2\pi, -4\pi}_{\times} \dots$$



$$f(x) = f(x + \pi)$$

$$\tan x = \tan(\pi + x) \quad \checkmark$$

$$T = (\pi), 2\pi, 3\pi, 4\pi, \dots$$

$\searrow$
 π



PERIODIC FUNCTIONS



Function $f(x)$ is called periodic function if :

$$f(x + T) = f(x) \quad \forall x \in D_f$$

Where : ('T' is a positive number)

Note : Smallest value of 'T' is called fundamental/principle period.

Graphically : Smallest length of graph along X-axis, which repeats itself.

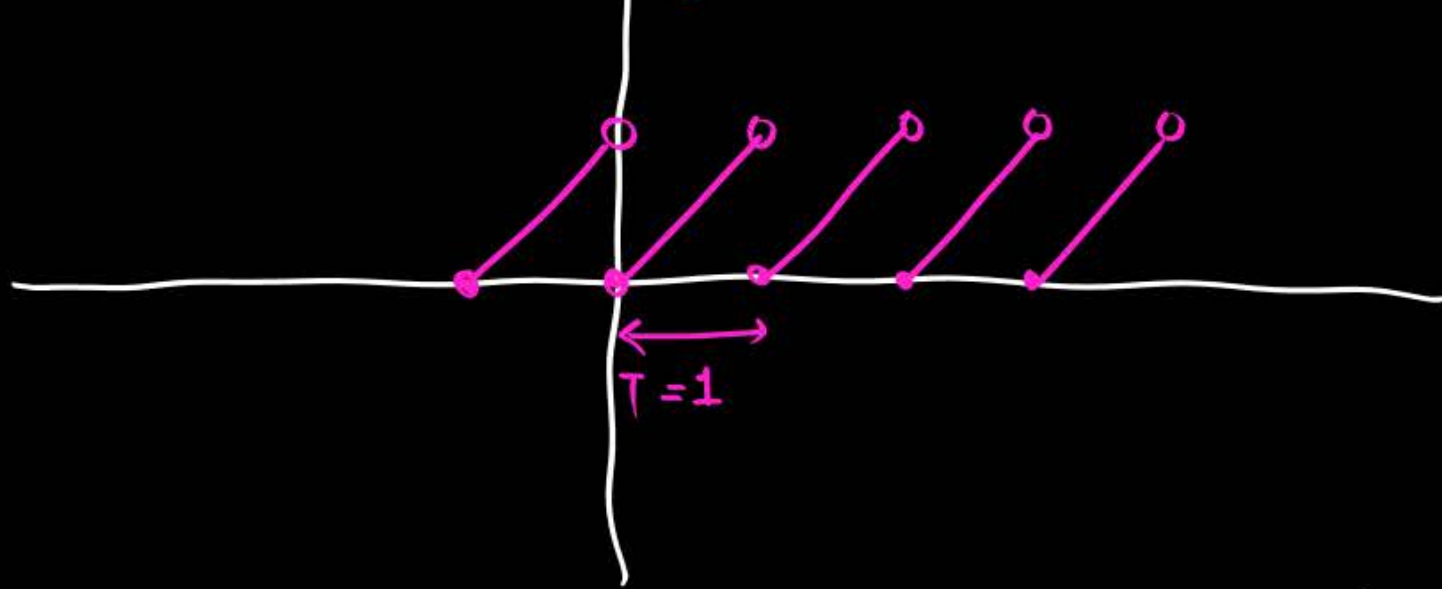


REMEMBER AS RESULT



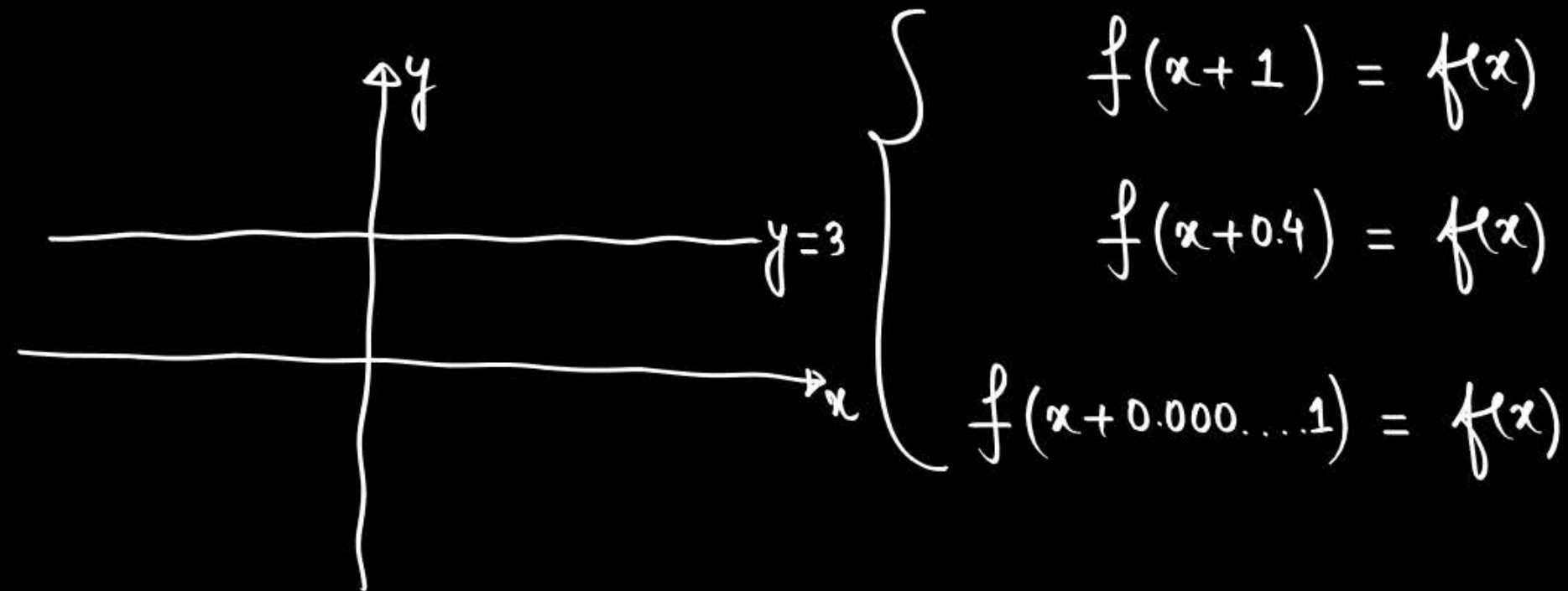
Functions	Periods
$\sin^n x, \cos^n x$ $\sec^n x, \operatorname{cosec}^n x$	π , if n is even 2π , if n is odd
$\tan^n x, \cot^n x$	π , n is even or odd
$ \sin x , \cos x , \tan x $ $ \cot x , \sec x , \operatorname{cosec} x $	π
$\{x\}$	1
Algebraic functions ex: $\sqrt{x}, x^2, x^3 + 5, \dots$ etc	Non-Periodic or aperiodic
$f(x) = \text{constant}$	“Periodic” with NO fundamental period

$$y = \{x\}$$



$$\# \sim f(x) = 3$$

$$\sim f(x+T) = f(x)$$





PROPERTIES OF PERIODIC FUNCTIONS

- ✓ 1. If 'T' is period of $f(x)$ then $2T, 3T, 4T, \dots, nT$ will also be period of $f(x)$.

$$f(x + nT) = f(x)$$

⇒ But fundamental period is only smallest one.

- ✓ 2. Periodic functions are non-invertible.

- ✓ 3. If $f(x)$ is periodic with period 'T' then :

(i) Period of : $\frac{1}{f(x)}, \sqrt{f(x)}, \alpha f(x) + \beta, (f(x))^2, \dots$ } $\textcircled{T}$

(ii) Period of : $f(ax + b) \Rightarrow \frac{T}{|a|}$

Period of : $\sin(3x - 8)$ is

Period of : $3\{7 - 4x\} + 11$ is

$\frac{2\pi}{3}$

$\frac{1}{|-4|} = \frac{1}{4}$



PROPERTIES OF PERIODIC FUNCTIONS

4. If period of $f(x)$ is T_1 & $g(x)$ is T_2 then :

Period of $f(x) \pm g(x)$, $\frac{f(x)}{g(x)}$, $f(x) \cdot g(x)$ is **"L.C.M. (T_1, T_2)"**

Note : Information about LCM.

If $\text{LCM}(a, b) = c \Rightarrow \left(\frac{c}{a} \& \frac{c}{b}\right)$ both should be "integer"

Otherwise LCM doesn't exist

$\text{LCM}\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{\text{LCM}(a, c, e)}{\text{HCF}(b, d, f)}$

DO YOU KNOW, HOW TO FIND LCM??



$$\text{LCM}(2, 8) = 8 \quad \checkmark$$

$$\text{LCM}(3\pi, 4\pi) = 12\pi \quad \checkmark$$

$$\text{LCM}(\sqrt{3}, 3) = 3 \quad \times$$

$\hookrightarrow = \text{dne.}$

$$\text{LCM}(2, \pi) = 2\pi \quad \times$$

$= \text{dne.}$

$$\text{LCM}(\sqrt{2}, \sqrt{8}) = 2\sqrt{2} \quad \checkmark$$

$$\text{LCM}(\pi, \pi^2) = \pi^2 \quad \times$$

$= \text{dne.}$

$$\text{LCM}(\sqrt{2}, \sqrt{3}) = \sqrt{6} \quad \times$$

$= \text{dne.}$

$$\text{LCM}\left(\frac{2}{3}, \frac{8}{4}, \frac{12}{8}\right) = \frac{24}{1}$$

NOTE:

$$\boxed{\text{LCM}(a, b) = c \checkmark}$$

→

$$\frac{c}{a} + \frac{c}{b} = \text{integer} \checkmark$$

= dne.

$$\boxed{\text{LCM}(a, \bar{a}) = \text{dne.}}$$



PROPERTIES OF PERIODIC FUNCTIONS

5. Period of composite function $f(g(x))$:

Case-I:

$g(x)$ is periodic with period 'T'

$f(x)$ is any function

$$\left. \begin{array}{l} \text{any func?} \\ \end{array} \right\} f(g(x)) = \text{periodic with period } \textcircled{T}$$

Case-II:

$g(x)$ is a linear $\Rightarrow g(x) = ax + b$

$f(x)$ is periodic function with period 'T'.

$$f(g(x)) = f(ax + b) \Rightarrow \textcircled{\frac{T}{|a|}}$$

Case-III:

$g(x)$ is non-linear/non-periodic

$f(x)$ is periodic with period 'T'

$$f(g(x)) = \text{non-periodic / aperiodic}$$

BASIC EXAMPLES



~~(i)~~ $e^{\{x\}}$ $\rightarrow$ $T=1$

~~(ii)~~ $\tan(\sin x)$ $\rightarrow$ $T=2\pi$

(iii) $\sin\left(\frac{1}{3x+2}\right)$ $\rightarrow$ ap.

(iv) $\{x^2 - 5x + 6\}$ $\rightarrow$ ap.

(v) $\sin\sqrt{x}$ $\rightarrow$ ap.

(vi) $\cos(x^2)$ $\rightarrow$ ap.



6. *Yaha sirf ek hi rule hai, ki yaha koi rule nhi hai*

Example :

Period of $f(x) = \cos(\sin x)$ $\rightarrow T = 2\pi, \pi$

Period of $f(x) = \cos(\sin x) + \cos(\cos x)$ $\rightarrow 2\pi, \pi, \frac{\pi}{2}$

Period of $f(x) = \underbrace{|\sin x|}_{\pi} + \underbrace{|\cos x|}_{\pi} \rightarrow \pi, \frac{\pi}{2}$ ✓

S³ ⇒ Options mein present, usse chote sabhi options ko check kre!

$$f(x)$$

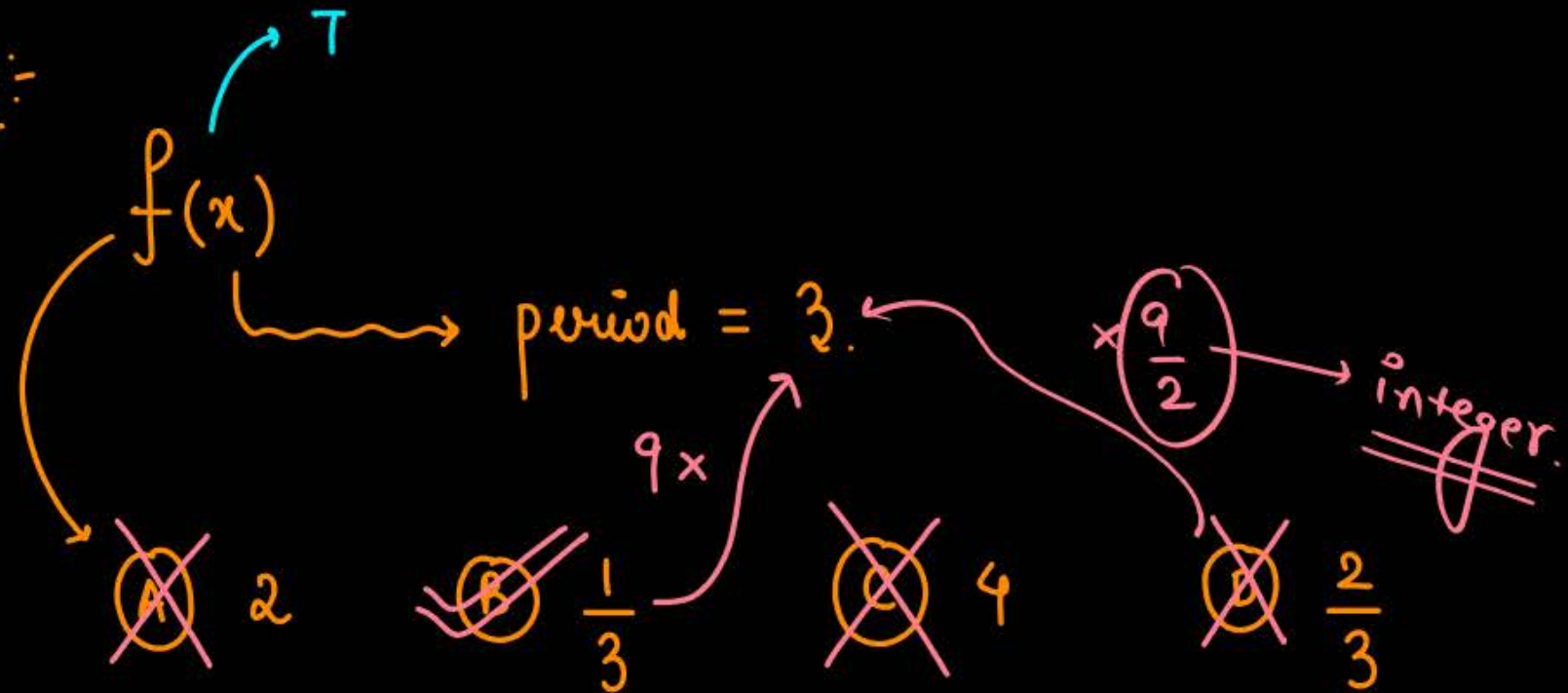


Rule & period = T

- $T_{\frac{1}{2}}, T_{\frac{1}{3}}, T_{\frac{1}{4}} \dots$ Check!

- Options mein T se small saare options check kro.

Ques:-



Bumper Question



Find period of following functions:

(i) $f(x) = \cos \frac{2x}{3} - \sin \frac{4x}{5}$ $\text{LCM}\left(\frac{3\pi}{1}, \frac{5\pi}{2}\right) = \frac{15\pi}{1} = 15\pi, \frac{15\pi}{2}, \frac{15\pi}{3}$

(ii) $f(x) = \sqrt{\sin 2x}$ $\frac{2\pi}{2} = \pi, \frac{\pi}{2}$

(iii) $f(x) = \sin^2 x + \cos^2 x$ $(\pi, \pi) \rightarrow T = \pi$ $\times$

(iv) $f(x) = \sec^2 x - \tan^2 x$ $\textcircled{3} f(x) = 1 \quad \forall x \in \mathbb{R}$
 periodic with f.p. is π

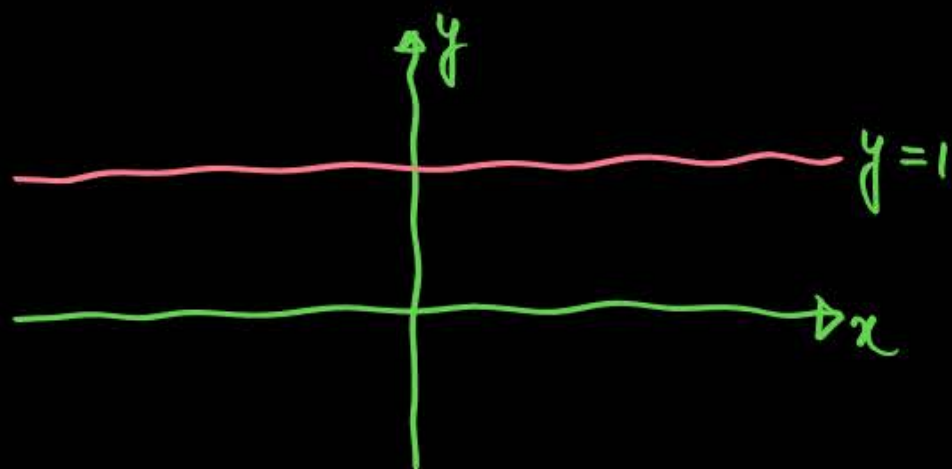
(v) $f(x) = \cos(3\pi x) + \left\{\frac{2x}{5} - 1\right\}$

(vi) $f(x) = \cos(x + \sin x)$

(vii) $f(x) = \cos^2 x + \sin^4 x$

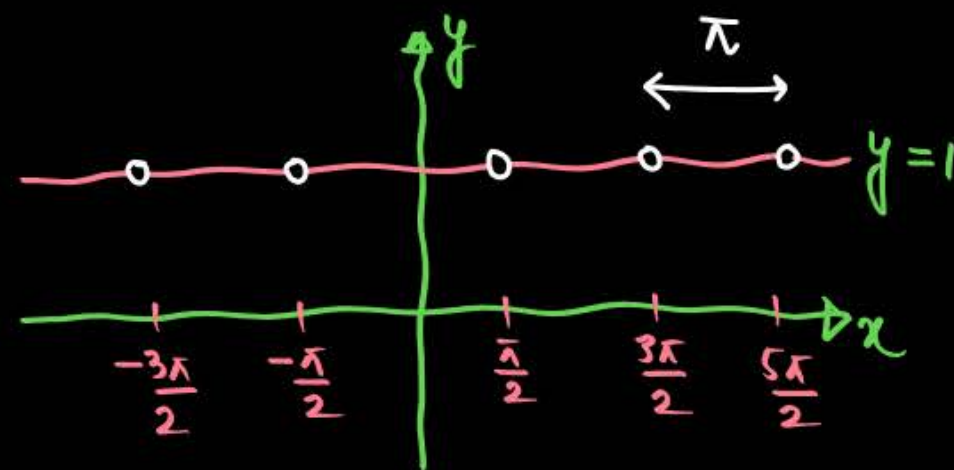
$\textcircled{4} f(x) = 1 \quad \forall x \in \mathbb{R} - \left\{(2n+1)\frac{\pi}{2}\right\}$

③ $f(x) = 1 \rightarrow \underline{\text{periodic}}$



④

$f(x) = 1, x \neq \text{odd multip. of } \frac{\pi}{2}$



$T = \pi \checkmark$

Bumper Question



Find period of following functions:

~~(i)~~ $f(x) = \cos \frac{2x}{3} - \sin \frac{4x}{5}$

~~(ii)~~ $f(x) = \sqrt{\sin 2x}$

~~(iii)~~ $f(x) = \sin^2 x + \cos^2 x$

~~(iv)~~ $f(x) = \sec^2 x - \tan^2 x$

~~(v)~~ $f(x) = \cos(3\pi x) + \left\{ \frac{2x}{5} - 1 \right\}$

~~(vi)~~ $f(x) = \cos(x + \sin x)$

~~(vii)~~ $f(x) = \cos^2 x + \sin^4 x$

(v) $\text{LCM} \left(\frac{2}{3}, \frac{5}{2} \right) = \frac{10}{1} = 10$

* (v) $f(x) = \underbrace{\cos x}_{\text{LCM}(2\pi, 1) = \text{dne.}} + \underbrace{\{x\}}_{\text{aperiodic}} \rightarrow \text{aperiodic.}$

$\text{LCM}(2\pi, 1) = \text{dne.}$

*** (vi) $f(x) = \cos(x + \sin x)$ $\rightarrow$ non-linear non-periodic.
 $\rightarrow$ a periodic x
 $\rightarrow$ period $= 2\pi$ ✓
 $x \rightarrow 2\pi + x$

$f(2\pi + x) = \cos(2\pi + x + \sin(2\pi + x))$
 $= \cos(x + \sin x) = f(x)$

$\frac{1}{2/5} = \frac{5}{2}$

$\frac{2\pi}{3\pi} = \frac{2}{3}$

(vi)

$$f(x) = \underbrace{\cos^2 x}_{\pi} + \underbrace{\sin^4 x}_{\pi} \longrightarrow T = \pi, \frac{\pi}{2}$$

$$f\left(\frac{\pi}{2} + x\right) = \cos^2\left(\frac{\pi}{2} + x\right) + \sin^4\left(\frac{\pi}{2} + x\right)$$

$$= \sin^2 x + \cos^4 x$$

$$= \sin^2 x + (1 - \sin^2 x)^2$$

$$= \sin^2 x + 1 + \sin^4 x - 2\sin^2 x = \cos^2 x + \sin^4 x = f(x)$$

Bumper Question



Find period of following functions:

(viii) $f(x) = \frac{\sin x + \sin 3x}{\cos x + \cos 3x}$

(ix) Graph of $f(x)$ symmetrical about line $x = 2$ & $x = 8$

(x) $[x] + [2x] + [3x] + \dots + [nx] - \frac{n(n+1)}{2}x$

(xi) $f(x) = 5\sin(2\sqrt{2}x) + 7\cos(3\sqrt{2}x)$

(xii) $f(x) = \frac{|\sin x + \cos x|}{|\sin x| + |\cos x|}$ HLW

Handwritten notes for (x):
 $x - \{x\} + 2x - \{2x\} + 3x - \{3x\} + \dots - \frac{n(n+1)}{2}x$
 $(x + 2x + 3x + \dots + nx) - (\{x\} + \{2x\} + \{3x\} + \dots)$
 $\text{LCM}\left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}\right) = \frac{1}{1} = 1$



(viii)

$$f(x) = \frac{\overset{D}{\sin x} + \overset{C}{\sin^3 x}}{\cos x + \cos 3x},$$

$$\cos x + \cos 3x \neq 0$$

$$2 \cos 2x \cdot \cos x \neq 0.$$

$$= \frac{\cancel{2} \sin 2x \cancel{\cos x}}{\cancel{2} \cos 2x \cdot \cancel{\cos x}}$$

$$x \neq (2n+1)\frac{\pi}{4}, (2m+1)\frac{\pi}{2}$$

$$f(x) = \tan 2x$$

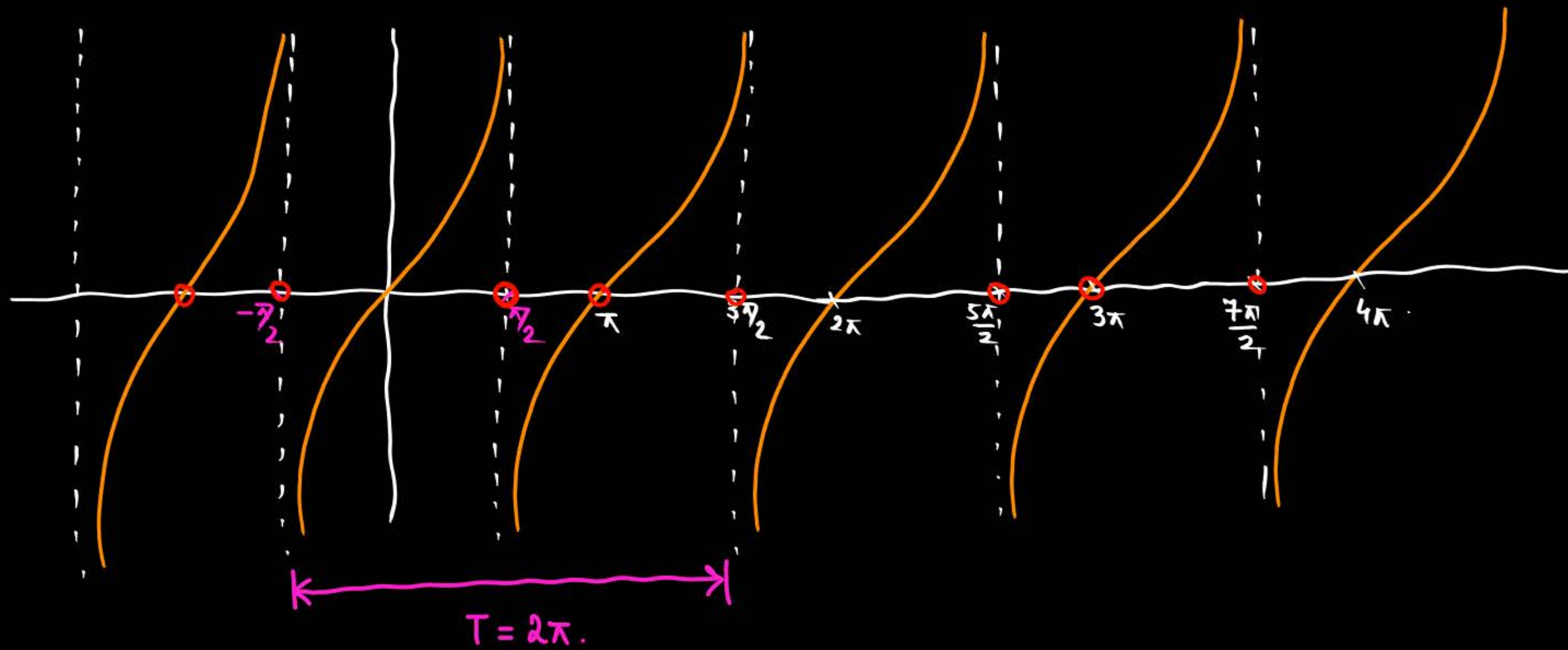
$\left(\frac{\pi}{2}\right) \times \left(2\pi\right)$

$$f(x) = \tan 2x$$

$$2x = 0$$

$$f(x) = \tan \theta, \quad \theta = (2n+1)\frac{\pi}{2}, (2m+1)\pi$$

$$\rightarrow 2\pi$$



9th

$$f(x) \xrightarrow{x=2} \boxed{f(2+x) = f(2-x)} \quad \text{--- (1)}$$

$$f(x) \xrightarrow{x=8} \boxed{f(8+x) = f(8-x)} \quad \text{--- (2)}$$

$$\boxed{f(2+x) = f(2-x)}$$

$$x \rightarrow x-6$$

$$f(2+x-6) = f(2-(x-6))$$

$$f(-4+x) = f(8-x)$$

$$f(x+8) = f(x-4)$$

$$x \rightarrow x+4$$

$$f(x+12) = f(x)$$

$$T=12$$

$$f(x+??) = f(\underline{x})$$

(T)

9th

$f(x)$

$x=2 \rightarrow f(2+x) = f(2-x) \quad \text{--- (1)}$

$x=8 \rightarrow f(8+x) = f(8-x) \quad \text{--- (2)}$

$x \rightarrow x+6$

$x \rightarrow 6-x$

$f(x+??) = f(x) \quad \text{--- (T)}$

$$f(14+x) = f(8-(x+6))$$

$$f(x+14) = f(2-x)$$

$$f(x+14) = f(x+2)$$

$$x \rightarrow x-2$$

$$f(x+12) = f(x)$$

$$f(14-x) = f(2+x)$$

$$f(14-x) = f(2-x)$$

$$x \rightarrow -x$$

$$f(14+x) = f(2+x)$$

Ques:



$$f(x) = |\sin x| + |\cos x|$$

$\pi, \frac{\pi}{2} \checkmark$

Smallest $x \in [0, \frac{\pi}{2}]$

$$\begin{aligned} f(x) &= \sin x + \cos x \in [1, \sqrt{2}] \quad \text{for } f \\ &= \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) \\ &\quad \downarrow \\ &\sqrt{2} \left[\frac{1}{\sqrt{2}}, 1 \right] \end{aligned}$$

$$\int_0^{4\pi} (|\sin x| + |\cos x|) dx$$

$$\xrightarrow{8(\frac{\pi}{2})} \int_0^{\frac{\pi}{2}} (|\sin x| + |\cos x|) dx$$

$$8 \int_0^{\frac{\pi}{2}} (\sin x + \cos x) dx$$

Critical Thinking Question (CTQ)



Let $f(x)$ be a function such that; $f(x-1) + f(x+1) = \sqrt{3} f(x)$, $\forall x \in \mathbb{R}$.

If $f(5) = 100$, find $\sum_{r=0}^{99} f(5 + 12r)$.

$$x \rightarrow x+1$$

$$f(x) + f(x+2) = \sqrt{3} f(x+1) \quad \text{--- (1)}$$

$$x \rightarrow x-1$$

$$f(x-2) + f(x) = \sqrt{3} f(x-1) \quad \text{--- (2)}$$

add.

$$2f(x) + f(x-2) + f(x+2) = \sqrt{3} (f(x+1) + f(x-1))$$

$$\begin{aligned} f(x-2) + f(x+2) &= 3f(x) - 2f(x) \\ &= f(x) \end{aligned}$$

$$\begin{aligned} f(5) + f(5+12) + f(5+12 \times 2) + \dots \\ \downarrow \quad \downarrow \quad \downarrow \\ 100 + 100 + 100 + \dots = 100 \times 100 = 10^4 \end{aligned}$$

$$f(x+12) = f(x)$$

$$f(x-2) + f(x+2) = f(x)$$

$$x \rightarrow x-2$$

$$f(x-4) + f(x) = f(x-2)$$

add

$$\cancel{f(x-2)} + f(x+2)$$

$$+ \cancel{f(x-4)} + \cancel{f(x)} = \cancel{f(x)} + \cancel{f(x-2)}$$

$$-f(x+2) = f(x-4)$$

$$x \rightarrow x+6$$

$$f(x+8) = -f(x+2)$$

$$f(x+8) = f(x-4)$$

$$x \rightarrow x+4$$

$$f(x+12) = f(x)$$

$$T=12$$

$$f(x+\text{cloud}) = f(x)$$

DO IT BY YOURSELF (DIBY)

DIBY-198

Let $f(x) = \frac{ax}{x+1}$, $x \neq -1$. Then, for what value of a is $f(f(x)) = x$? [IIT-JEE 2001]

- (A) $-\sqrt{2}$ (B) $\sqrt{2}$ (C) 1 (D) -1

[Ans.: (D)]

DIBY-199

If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then $g\{f(x)\}$ is invertible in the domain [IIT-JEE 2004]

- (A) $\left[0, \frac{\pi}{2}\right]$ (B) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ (C) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (D) $[0, \pi]$

[Ans.: (B)]

DIBY-200

Let function $f: R \rightarrow R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then, f is [IIT-JEE 2002]

- (A) one-to-one and onto (B) one-to-one but not onto
(C) onto but not one-to-one (D) neither one-to-one nor onto

[Ans.: (A)]

DIBY-201

There are exactly two distinct linear functions _____ and _____ which map $\{-1, 1\}$ on to $\{0, 2\}$. **[IIT-JEE 1989]**

[Ans. : $f(x) = \pm x + 1$]

DIBY-202

Consider the following functions in their default domains:

- (i) $\frac{2^x-1}{2^x+1}$ (ii) $\frac{x^2+1}{x \sin x}$ (iii) $\ln \left(\frac{1+x}{1-x} \right)$ (iv) $xe^{|x|+\cos x}$

Which of these is/are odd:

- (A) (i) & (iii) (B) (i) & (iv) (C) All four (D) (i), (iii) & (iv)

[Ans. (D)]

DIBY-203

Let $f: [-4, 4] \sim \{-\pi, 0, \pi\} \rightarrow \mathbf{R}$ such that $f(x) = \cot(\sin x) + \left[\frac{x^2}{|a|} \right]$, where $[.]$ denotes the greatest integer function is an odd function, complete set of values of ' a ' is:

- (A) $(-16, 16) \sim \{0\}$ (B) $(-\infty, -16) \cup (16, \infty)$
 (C) $[-16, 16] \sim \{0\}$ (D) $(-\infty, -16) \cup [16, \infty)$

[Ans. (B)]

DIBY-204

Let $f(x)$ be an odd function defined on $\mathbb{R}$ such that $f(1) = 2$, $f(3) = 5$ and $f(-5) = -1$. The value of $\frac{f(f(f(-3))) + f(f(0))}{3f(1) - 2f(3) - f(5)}$ is:

(A) $-\frac{2}{5}$

(B) $-\frac{2}{3}$

(C) $\frac{2}{5}$

(D) 1

[Ans. (C)]

DIBY-205

f is an even function defined on the interval $(-5, 5)$, then four real values of x satisfying the equation $f(x) = f\left(\frac{x+1}{x+2}\right)$ are

[IIT-JEE 1996]

[Ans. : $x = \frac{\pm 3 \pm \sqrt{5}}{2}$]

DIBY-206

Let $f(x)$ be an odd function defined on $(-\infty, \infty)$ and $g(x)$ be an even function defined on $(-\infty, \infty)$. If $f(x) - g(x) = x^2 + 5x + 7$, then $g(2)$ is equal to

[Ans. -11]

DIBY-207

Let $f(x) = e^{\{e^{|x|}\operatorname{sgn} x\}}$ and $g(x) = e^{[e^{|x|}\operatorname{sgn} x]}$, $x \in \mathbb{R}$ where $\{x\}$ and $[]$ denotes the fractional part and integral part functions respectively.

Also, $h(x) = \ln(f(x)) + \ln(g(x))$, then for all real x , $h(x)$ is:

- (A) an odd function
- (B) an even function
- (C) neither an odd nor an even function
- (D) both odd as well as even function

[Ans. (A)]

DIBY-208

If $h(x) = A x^5 + B \sin x + C \ln \left(\frac{1+x}{1-x} \right) + 7$, where A, B, C are non-zero real constants

and $h\left(\frac{-1}{2}\right) = 6$, then find the value of $h\left(\frac{\operatorname{sgn}(e^{-x})}{2}\right)$.

[Ans. 8]

DIBY-209

Let $f(x) = a^x (a > 0)$ be written as $f(x) = f_1(x) + f_2(x)$, where $f_1(x)$ is an even function and $f_2(x)$ is an odd function. Then $f_1(x+y) + f_1(x-y)$ equals is:

- (A) $2f_1(x+y)f_2(x-y)$ (B) $2f_1(x)f_1(y)$
(C) $2f_1(x+y)f_1(x-y)$ (D) $2f_1(x)f_2(y)$

[Ans. (B)]

DIBY-210

Four statements, about the graph of $f(x) = \log_{10} \frac{1+x}{1-x}$, are given below.

Mark all the correct statements:

- I: the graph is symmetric about the y -axis.
II: the graph lies in the 4th and 1st quadrants.
III: the graph is symmetric about the origin.
IV: the graph lies in the 3rd and 1st quadrants and goes through the origin.

- (A) I and IV (B) II and III (C) III (D) III and IV

[Ans. (D)]

MISCELLANEOUS QUESTIONS



DIBY-211

If the function $f: (-\infty, -1] \rightarrow (a, b]$ defined by $f(x) = e^{x^3-3x+1}$ is one-one and onto, then the distance of the point $P(2b+4, a+2)$ from the line $x + e^{-3}y = 4$ is:

[JEE Mains-2024]

- (A) $2\sqrt{1+e^6}$ (B) $4\sqrt{1+e^6}$ (C) $3\sqrt{1+e^6}$ (D) $\sqrt{1+e^6}$

[Ans. (A)]

DIBY-212

Let $A = \{a_1, a_2, a_3, a_4\}$ & $B = \{b_1, b_2, b_3, b_4, b_5\}$. The number of injective functions f from A to B such that $f(a_1) \neq b_4$ and $f(a_2) = b_1$, is

[Ans. 18]

DIBY-213

Number of functions $f: \{1, 2, \dots, 100\} \rightarrow \{0, 1\}$, that assign 1 to exactly one of the positive integers less than or equal to 98, is equal to _____.

[JEE Mains-2025]

[Ans. 392]

MISCELLANEOUS QUESTIONS

DIBY-214

Consider the sets $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 = 25\}$,

$B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + 9y^2 = 144\}$,

$C = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 4\}$,

and $D = A \cap B$. The total number of one-one functions from the set D to the set C is:

(A) 15120

(B) 19320

(C) 17160

(D) 18290

[Ans. (c)]

DIBY-215

Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the number of one-one functions $f: S \rightarrow P(S)$, where $P(S)$ denote the power set of S , such that $f(n) \subset f(m)$ where $n < m$ is _____.

[JEE Mains-2023]

[Ans. 3240]

DIBY-216

Let a function f be defined as $f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$. If f satisfy $f(f(x)) = f(x)$, $\forall x \in \{1, 2, 3, 4\}$, the number of such functions is _____.

[Ans. 41]

MISCELLANEOUS QUESTIONS



DIBY-217

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6, 7\}$ are two sets. Let m is the number of one-one functions $f: A \rightarrow B$ such that $f(i) \neq i \forall i \in A$ and n is the number of one-one functions $f: A \rightarrow B$ such that $|f(i) - i| \leq 3 \forall i \in A$, then find the value of $(m + n)$.

[Ans. 94]

DIBY-218

If $f(x) = \sin^2 x + \sin^2 \left(x + \frac{\pi}{3}\right) + \cos x \cos \left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$, then find $(g \circ f)(x)$:

[IIT-JEE 1996]

[Ans. 1]

DIBY-219

Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X , then the value of $\frac{1}{5!}(\beta - \alpha)$ is ____.

[JEE (Adv.)-2018, Paper-2]

[Ans. 119]

DIBY-222

Find the period of the following function:

- (i) $f(x) = \operatorname{cosec}^2 3x + \cot 4x$
- (ii) $f(x) = \frac{1}{2} \left(\frac{|\sin x|}{\cos x} + \frac{\sin x}{|\cos x|} \right)$
- (iii) $f(x) = \tan \frac{\pi}{2} [x]$, where $[.]$ denotes greatest integer function
- (iv) $f(x) = |\cos \pi x| + |\cos 2\pi x| + \dots + |\cos n\pi x| + e^{x-[x]}$ where $[.]$ is greatest integer function.
- (v) $f(x) = \operatorname{sgn}([x] + [-x])$ where $[.]$ denotes greatest integer function
- (vi) $f(x) = \exp(x - [x] + \sqrt{x - [x]} + (x - [x])^2 + |\sin \pi x| + |\cos \pi x| + |\tan \pi x|)$ where $[.]$ is GIF
- (vii) $f(x) = \sin 3x \cos [3x] - \cos 3x \sin [3x]$, where $[.]$ denotes the greatest integer function

[Ans. : (i) π ; (ii) 2π ; (iii) 2; (iv) 1; (v) 1; (vi) 1; (vii) $1/3$]



Homework



Re-attempt all good Questions of Class

DIBY

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians